# Institut Universitaire des Sciences

**Faculté :** **Faculté Science et Technologie**

**TD** **N° : 2-**Système

**Prenoms et Nom :** Gougou Zachary **COMPER**

**Niveau :** Licence 3 (L3)

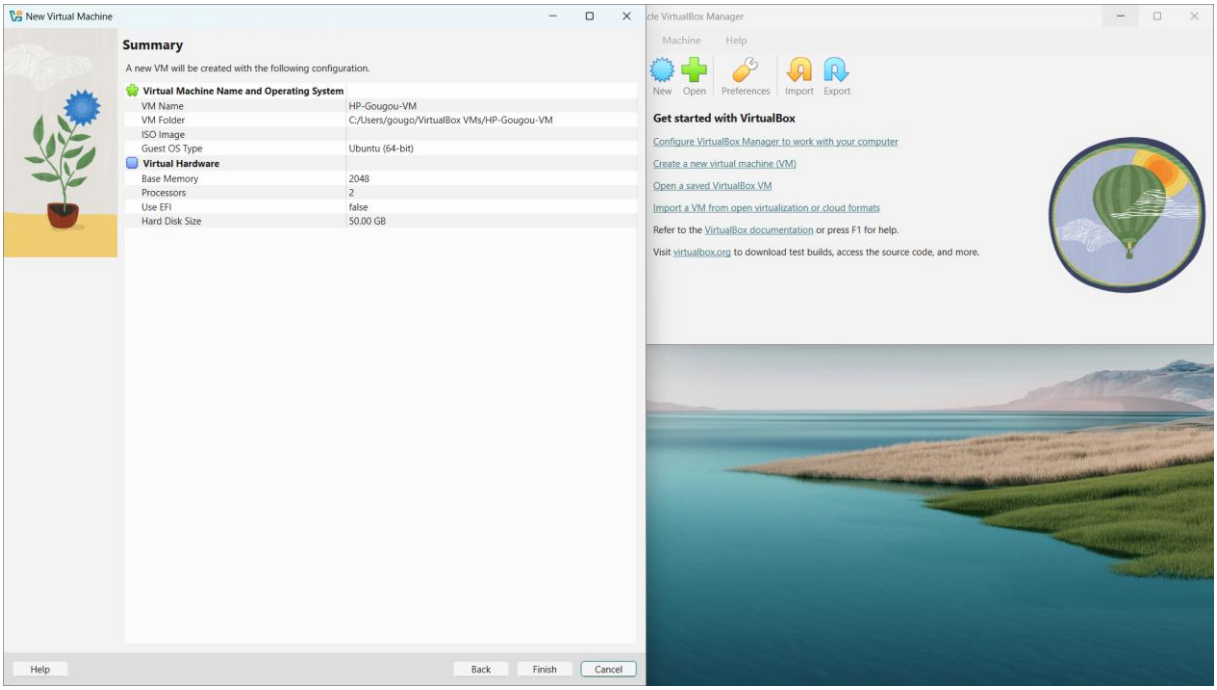
**Professeur** Ismael **SAINT AMOUR**

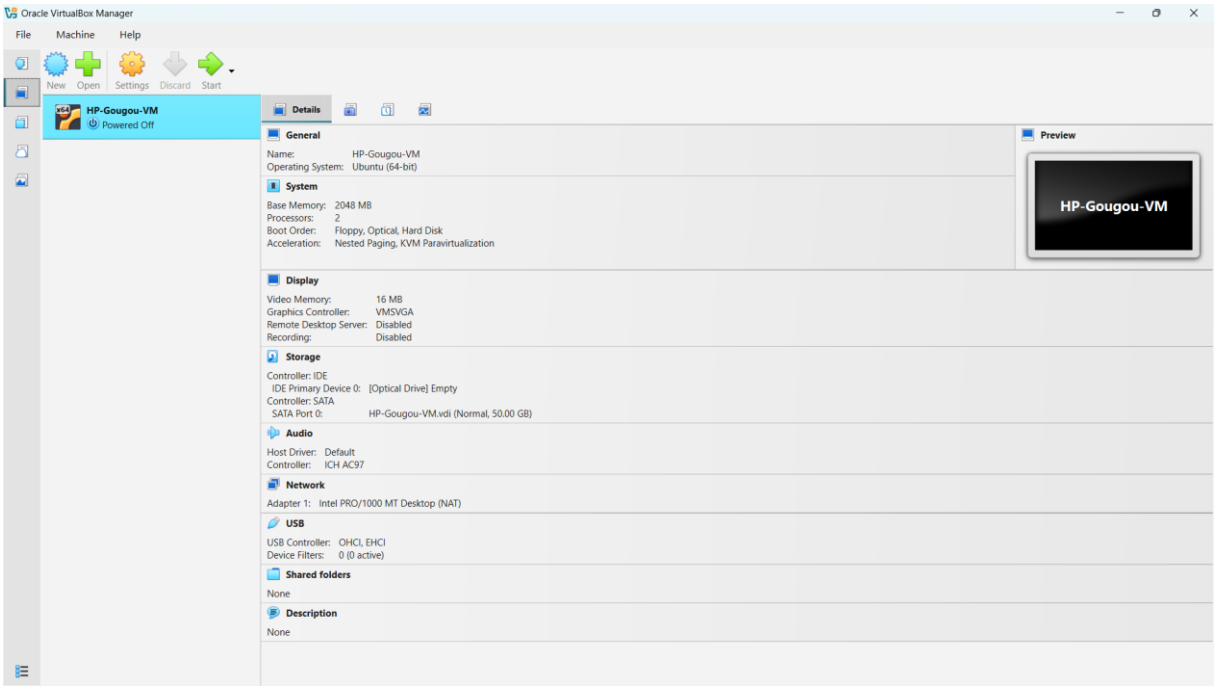
**Date :** Fevrier 2025

**TRAVAIL DIRIGE(LABORATOIRE)-SYSTEME**

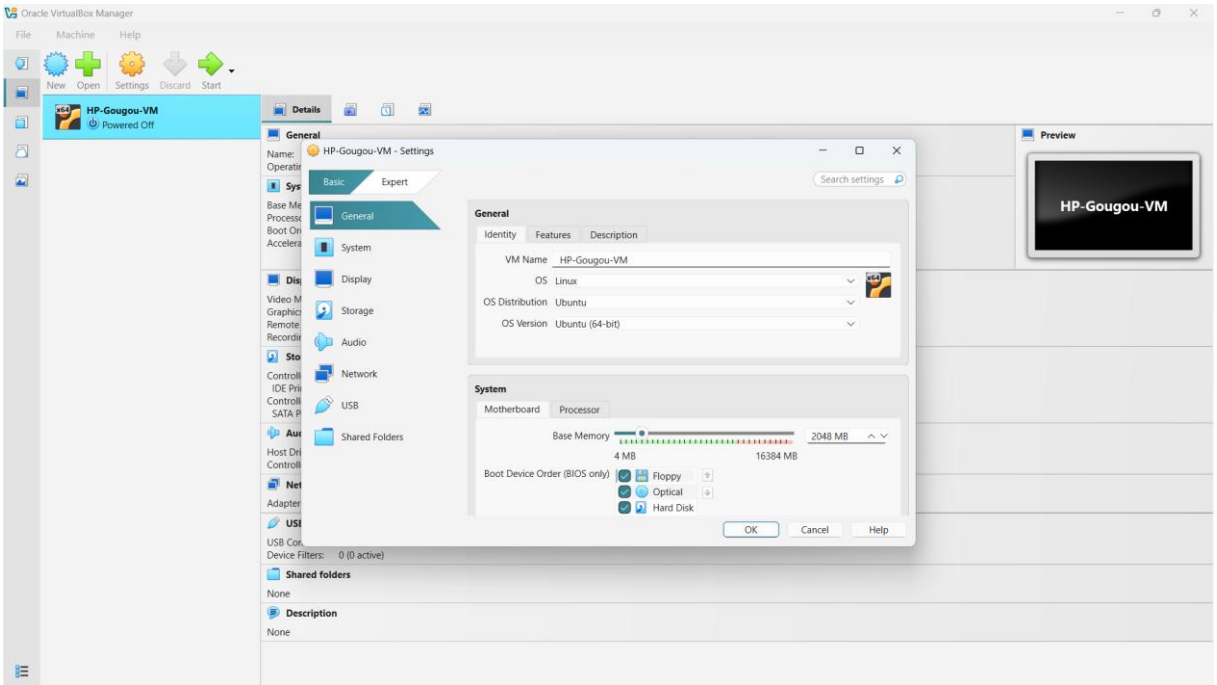
**OBJECTIF** : Ce deuxième travail dirigé est dédié à la mise en œuvre pratique de la virtualisation système à l'aide du logiciel Oracle VM VirtualBox. L'objectif principal est de maîtriser le déploiement complet d'un environnement Linux Ubuntu, depuis le provisionnement des ressources matérielles (RAM, CPU, stockage) et la configuration réseau, jusqu'à l'installation et la gestion opérationnelle de la machine virtuelle (VM). Ce projet permet ainsi d'assimiler les concepts fondamentaux de la virtualisation en environnement professionnel.

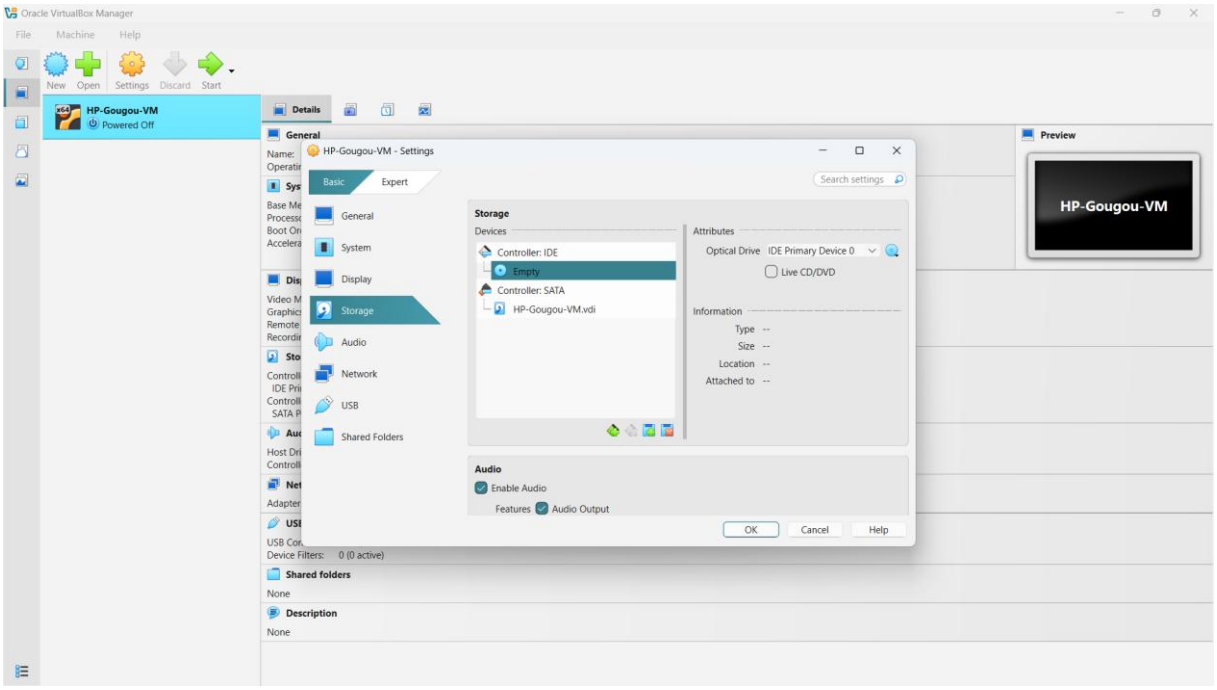
1-Creation d'une nouvelle machine virtuelle après avoir installé le Logiciel Virtual Box

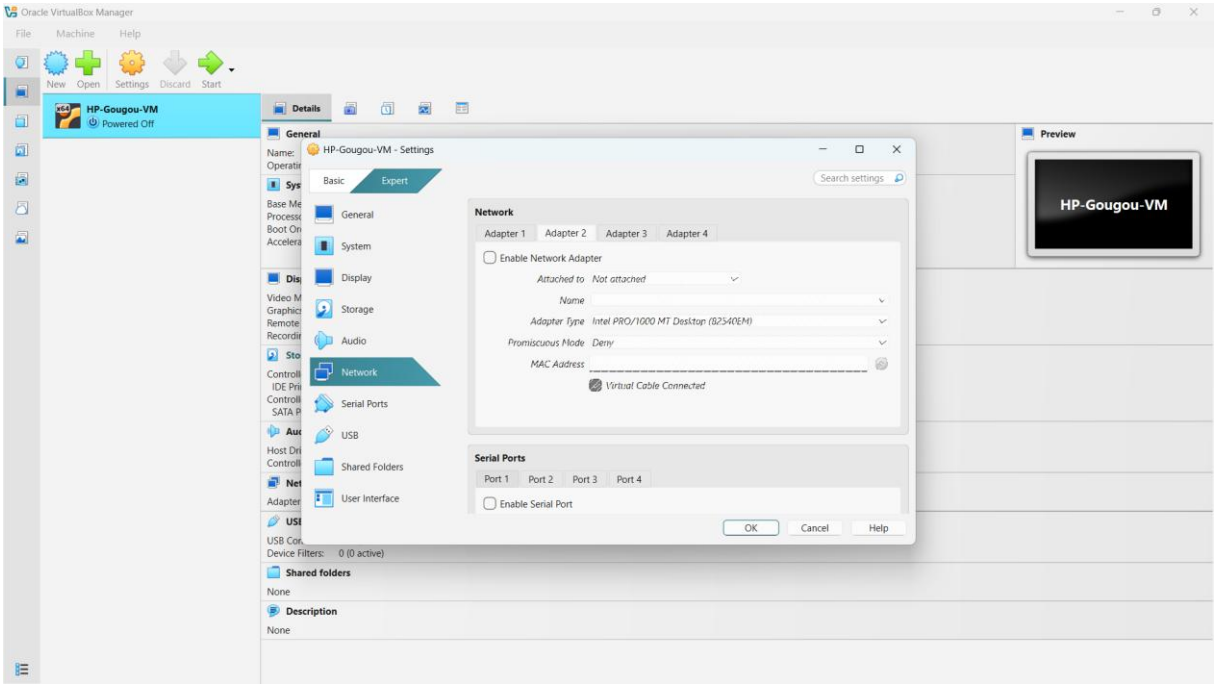




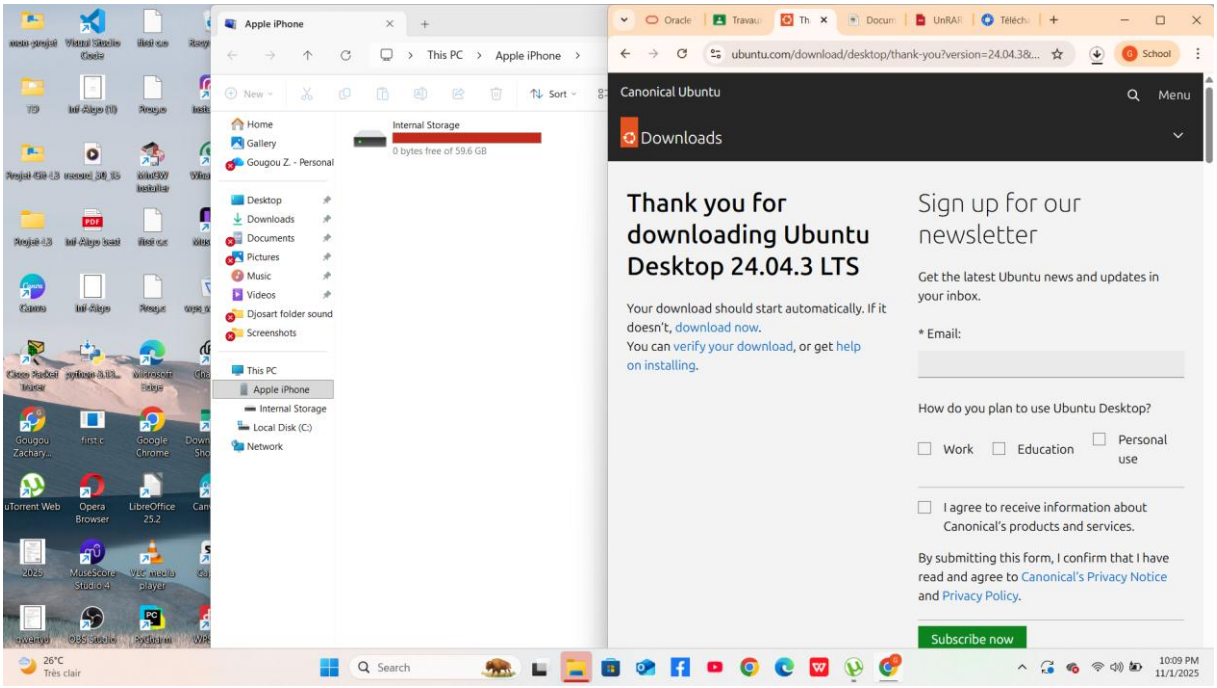

Configuration de la Machine(Système, Processeur, Affichage, Stockage,…)

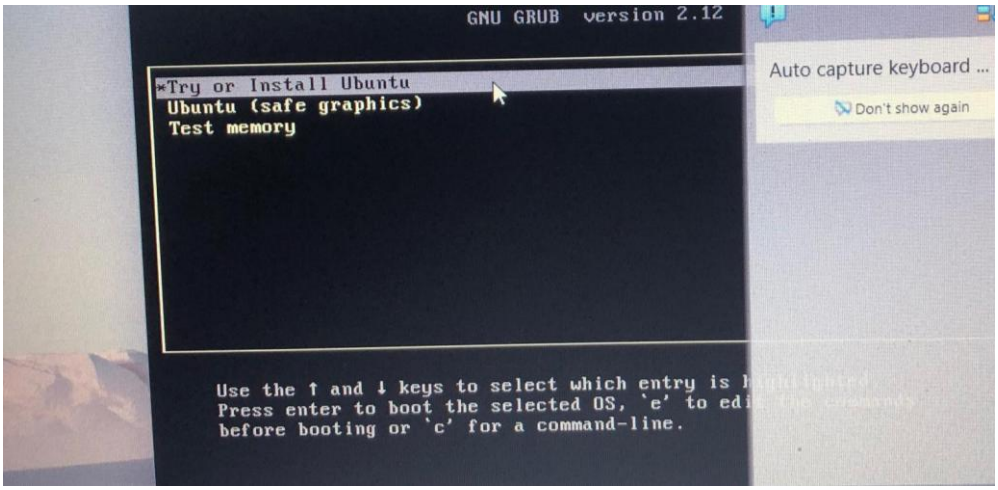


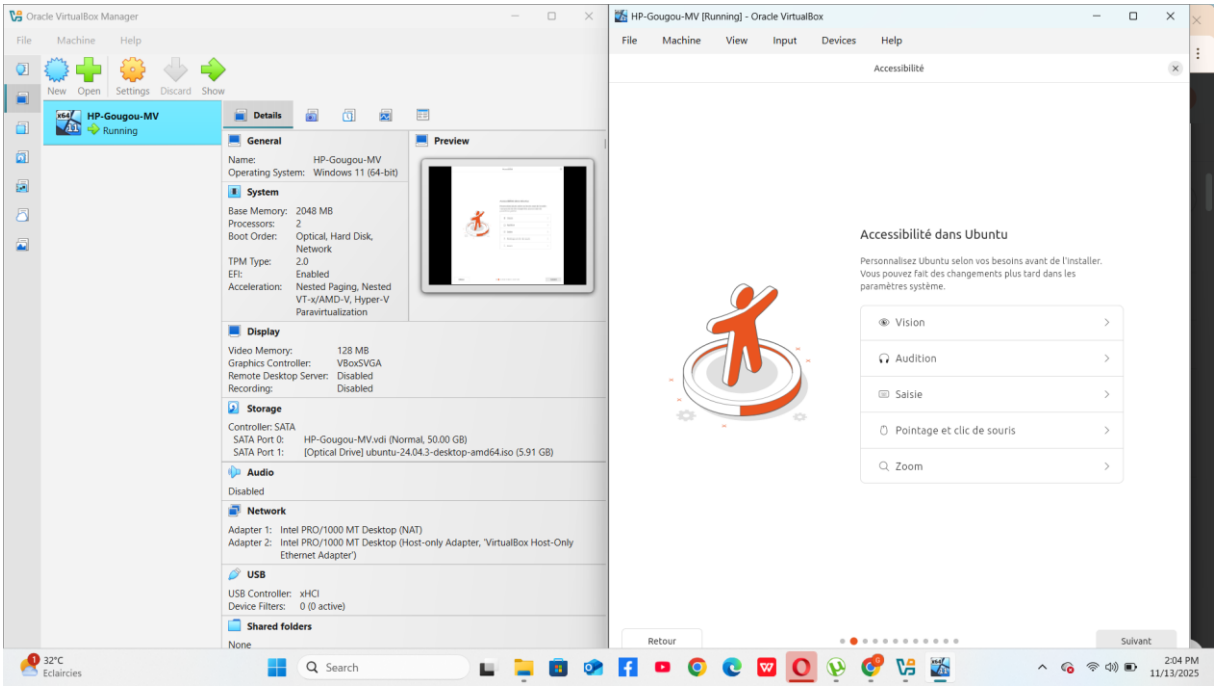

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
HP-Gougou-VM
Powered Off
Details
General
Name:
HP-Gougou-VM - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
USB
Shared Folders
Storage
Devices
Controller: IDE
Empty
Controller: SATA
HP-Gougou-VM.vdi
Attributes
Optical Drive
IDE Primary Device 0
Live CD/DVD
Information
Type --
Size --
Location --
Attached to --
Audio
Enable Audio
Features
Audio Output
OK
Cancel
Help
Device Filters: 0 (0 active)
Shared folders
None
Description
None
Preview
HP-Gougou-VM

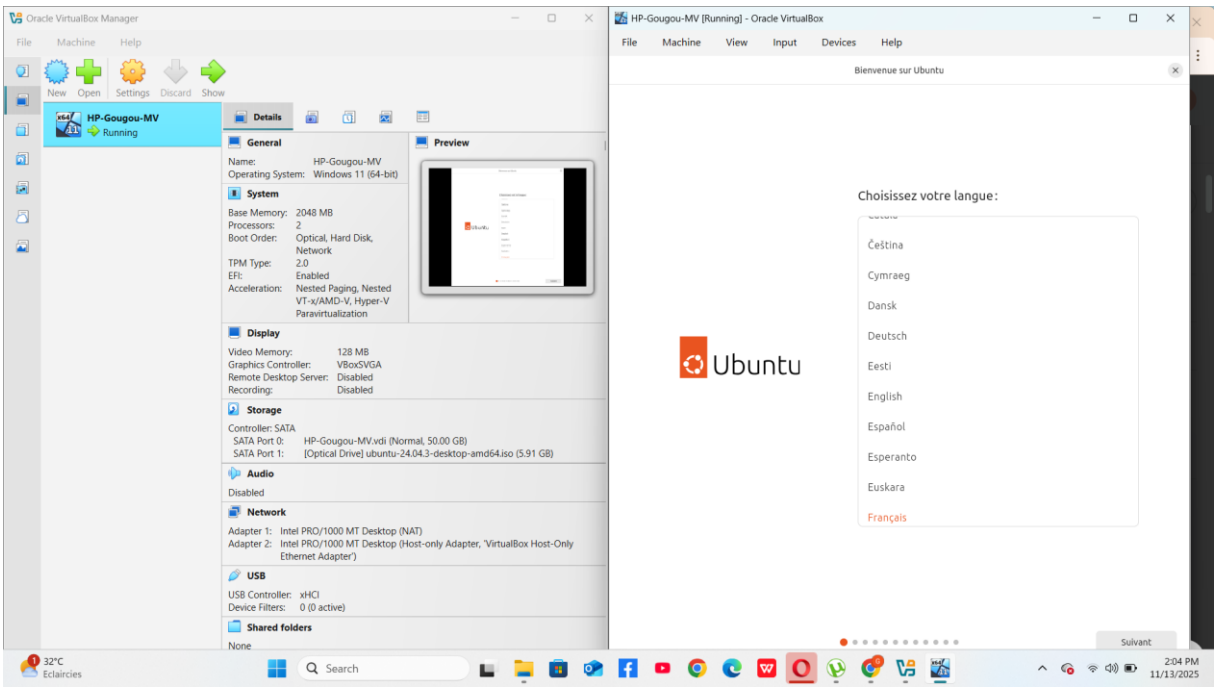

Oracle VirtualBox Manager
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General
Name:
HP-Gougou-VM - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
Not attached
Name
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
Device Filters: 0 (0 active)
Shared folders
None
Description
None
Preview
HP-Gougou-VM

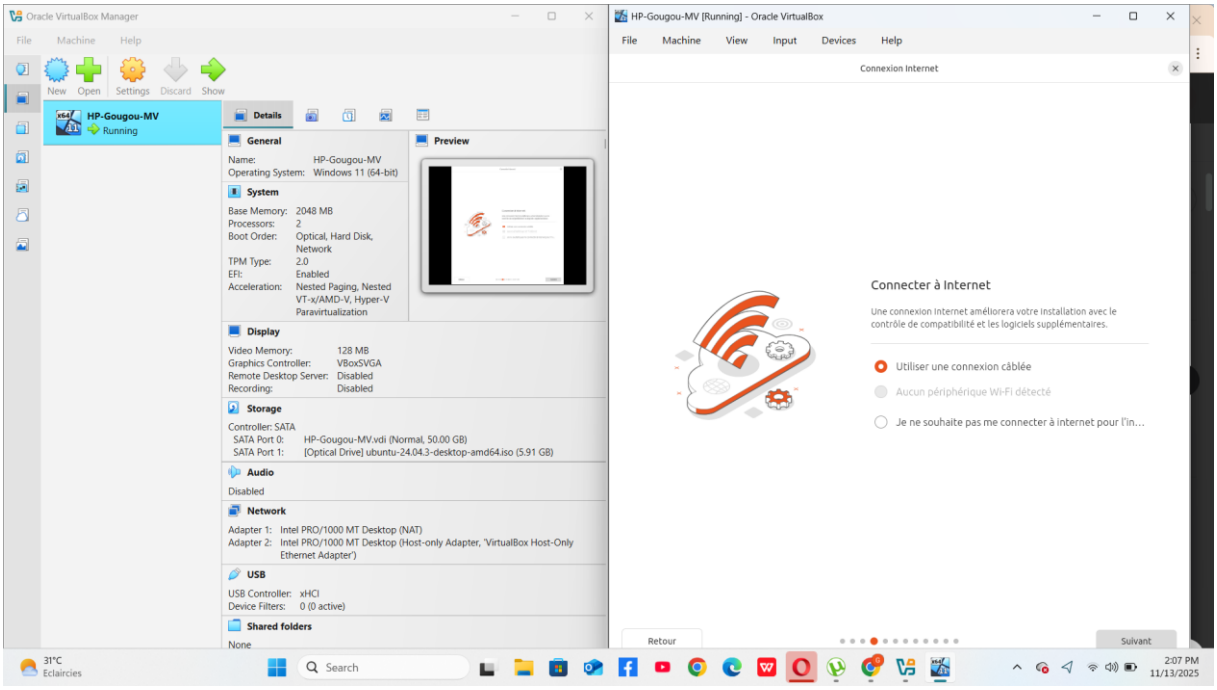
Installation d’Ubuntu, faut tout d’abord telecharger le fichier (ISO Ubuntu Server)

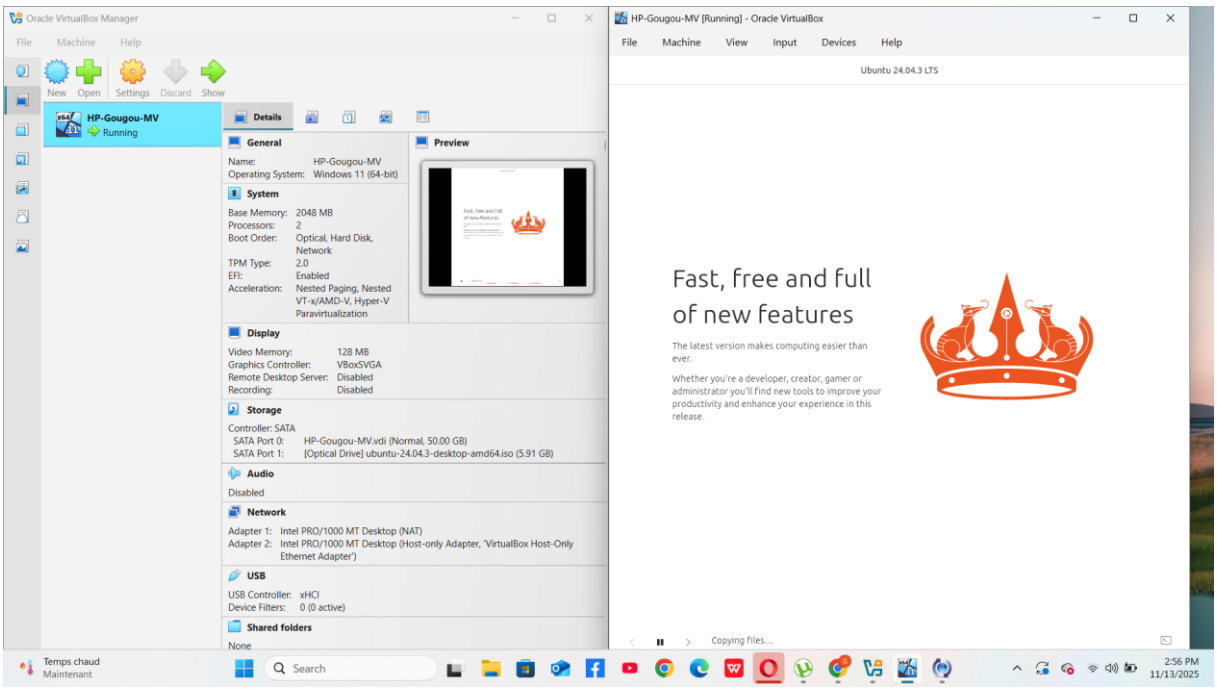


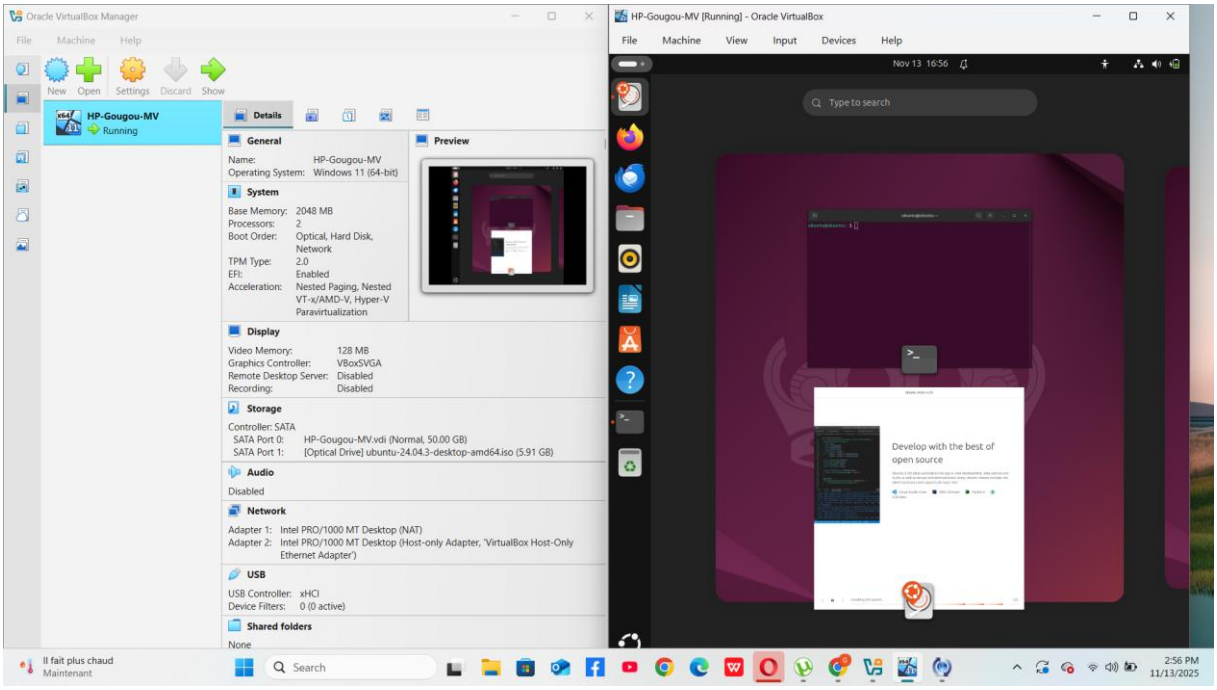



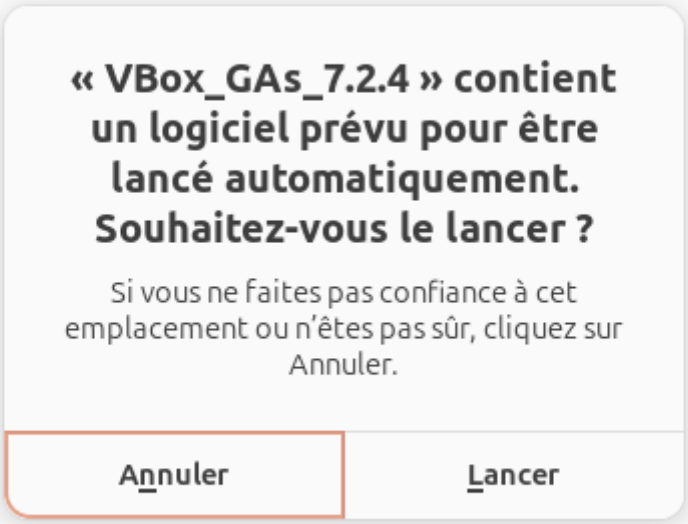

Oracle VirtualBox Manager
File Machine Help
New Open Settings Discard Show
HP-Gougou-MV
Running
Details
General
Name: HP-Gougou-MV
Operating System: Windows 11 (64-bit)
System
Base Memory: 2048 MB
Processors: 2
Boot Order: Optical, Hard Disk, Network
TPM Type: 2.0
EFI: Enabled
Acceleration: Nested Paging, Nested VT-x/AMD-V, Hyper-V Paravirtualization
Preview
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: HP-Gougou-MV.vdi (Normal, 50.00 GB)
SATA Port 1: [Optical Drive] ubuntu-24.04.3-desktop-amd64.iso (5.91 GB)
Audio
Disabled
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
Adapter 2: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')
USB
USB Controller: xHCI
Device Filters: 0 (0 active)
Shared folders
None
HP-Gougou-MV [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Accessibilité
Accessibilité dans Ubuntu
Personnalisez Ubuntu selon vos besoins avant de l'installer. Vous pouvez fait des changements plus tard dans les paramètres système.
Vision
Audition
Saisie
Pointage et clic de souris
Zoom
Retour
Suivant
32°C
Eclaircies
Search
2:04 PM
11/13/2025

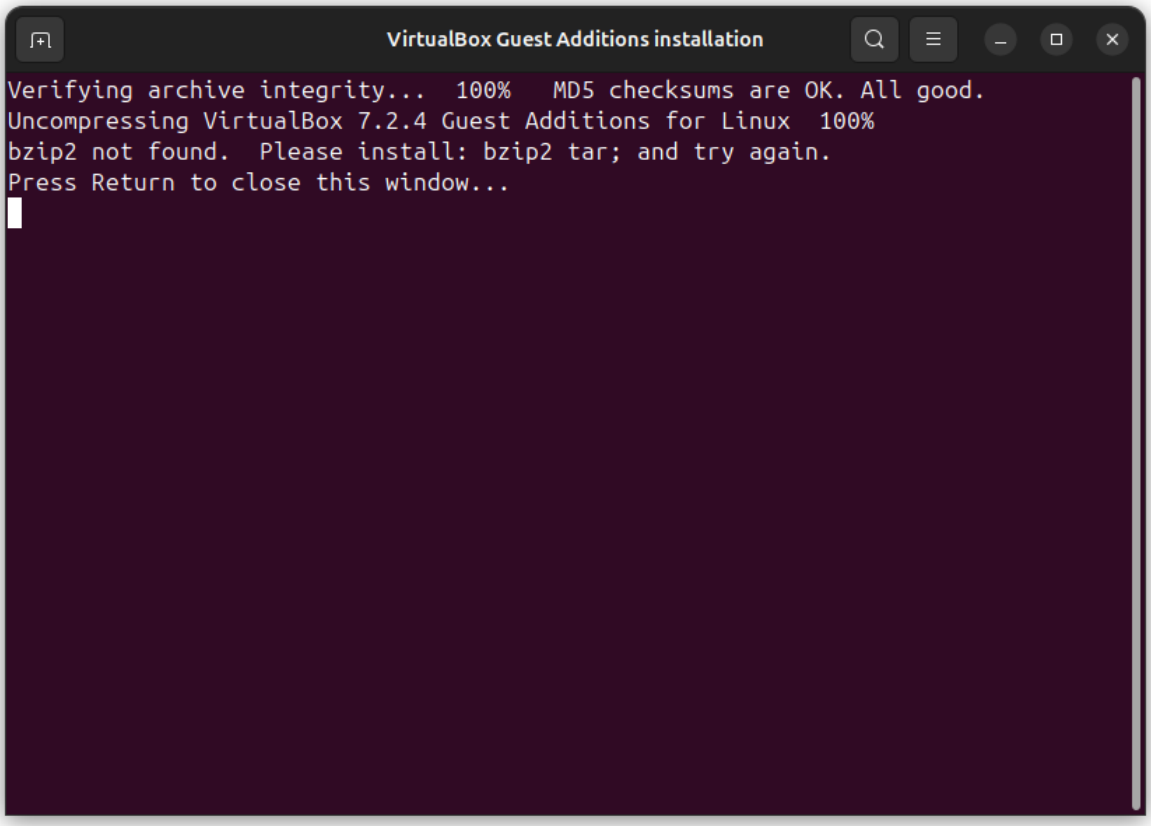

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Adapter 2: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')
USB
USB Controller: xHCI
Device Filters: 0 (0 active)
Shared folders
None
HP-Gougou-MV [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Bienvenue sur Ubuntu
Ubuntu
Choisissez votre langue :
Čeština
Cymraeg
Dansk
Deutsch
Eesti
English
Español
Esperanto
Euskara
Français
Suivant
32°C
Eclaircies
Search
2:04 PM
11/13/2025


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USB Controller: xHCI
Device Filters: 0 (0 active)
Shared folders
None
HP-Gougou-MV [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Connexion Internet
Connecter à Internet
Une connexion Internet améliorera votre installation avec le contrôle de compatibilité et les logiciels supplémentaires.
Utiliser une connexion câblée
Aucun périphérique Wi-Fi détecté
Je ne souhaite pas me connecter à internet pour l'in…
Retour
Suivant
31°C
Eclaircies
Search
2:07 PM
11/13/2025

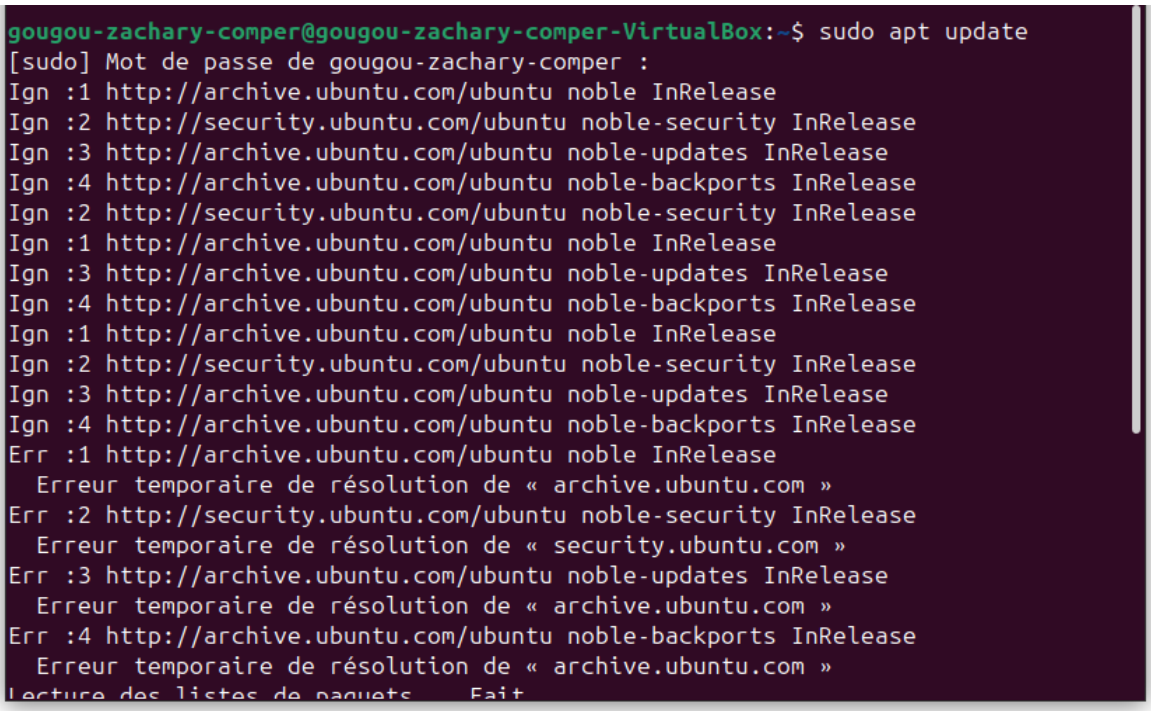

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Remote Desktop Server: Disabled
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USB Controller: xHCI
Device Filters: 0 (0 active)
Shared folders
None
HP-Gougou-MV [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Ubuntu 24.04.3 LTS
Fast, free and full of new features
The latest version makes computing easier than ever.
Whether you're a developer, creator, gamer or administrator you'll find new tools to improve your productivity and enhance your experience in this release.
Copying files…
Temps chaud
Maintenant
Search
2:56 PM
11/13/2025


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New Open Settings Discard Show
HP-Gougou-MV
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General
Name: HP-Gougou-MV
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USB Controller: xHCI
Device Filters: 0 (0 active)
Shared folders
None
HP-Gougou-MV [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Nov 13 16:56
Type to search
Develop with the best of open source
Il fait plus chaud
Maintenant
Search
2:56 PM
11/13/2025


« VBox_GAs_7.2.4 » contient un logiciel prévu pour être lancé automatiquement. Souhaitez-vous le lancer ?
Si vous ne faites pas confiance à cet emplacement ou n'êtes pas sûr, cliquez sur Annuler.
Annuler
Lancer




Mis a jour du Système avec les commandes : sudo apt update, sudo apt upgrade –y (via le Terminal)



Ce travail dirige(TD) m’a permis de  valider le déploiement complet d’un environnement virtualisé à travers trois phases essentielles : le provisionnement des ressources matérielles sous l'hyperviseur, l'installation guidée du système d'exploitation Ubuntu, et la mise à niveau logicielle post-installation. En conclusion, ce laboratoire a permis d'assimiler les étapes clés de la création d'un serveur. Nous disposons désormais d'une plateforme Linux stable, sécurisée et opérationnelle, constituant une base solide pour de futures configurations de services réseau.

Il y’avait presque pas de difficultés, c’est un TD assez simple dès qu’on dispose d’un Ordinateur d’une capacité assez efficace(RAM, CPU,…). L’essentiel, c’est de bien suivre les étapes.